

CSE5095: Types and Programming Languages

Homework 1: Inference rules, derivations, rule induction

Your name here

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1 Logistics

Important note: This document **DOES NOT** have all of the information you need to complete the assignment. It's just a template for typesetting your answers. You **MUST** read the writeup pdf.

Great to see you're (at least thinking about) writing up your solution in LaTeX though! Thanks! See HuskyCT for resources if you need help. Some useful macros are defined in the file `macros.tex`, and this source file gives examples of how to use them.

2 XML

Task 1. Add another inference rule for showing that the concatenation of 3 XML strings (e.g., XYZ) is also an XML string.

Answer:

$$\frac{X \text{ is XML} \quad Y \text{ is XML} \quad Z \text{ is XML}}{XYZ \text{ is XML}} \text{ (IDENTITYRULE)}$$

Task 2. Write a derivation for the conclusion XYZ is XML assuming the premises of the rule you wrote in Task 1. Make sure to include the names/numbers of the rules you're using.

Answer:

$$\frac{\frac{X \text{ is XML} \quad Y \text{ is XML}}{XY \text{ is XML}} \text{ (X-4)} \quad Z \text{ is XML}}{XYZ \text{ is XML}} \text{ (X-4)}$$

Typesetting Hint: you can typeset a derivation by nesting more inference rules (`\Rule`) in the premises of an inference rule. The code for this gets messy, so make sure you're using indentation or other hints to keep yourself sane. If you're already familiar with using LaTeX for math, you may be tempted to just use `\frac`, but this gets really difficult to read because it makes the nested fractions smaller and smaller (and also doesn't let you include the rule names).

3 Red-Black Trees

Code for the inference rules and definitions is below so you can have examples of how to typeset these.

$$\begin{aligned} \text{Trees } T &::= \text{Leaf} \mid \text{BNode}(T, T) \mid \text{RNode}(T, T) \\ \text{Colors } c &::= \text{Black} \mid \text{Red} \end{aligned}$$

$$\frac{}{\text{Leaf isTree Black } 1} \text{ (RBT-1)} \quad \frac{T_1 \text{ isTree } c_1 \ n \quad T_2 \text{ isTree } c_2 \ n}{\text{BNode}(T_1, T_2) \text{ isTree Black } n+1} \text{ (RBT-2)}$$

$$\frac{T_1 \text{ isTree Black } n \quad T_2 \text{ isTree Black } n}{\text{RNode}(T_1, T_2) \text{ isTree Red } n} \text{ (RBT-3)}$$

$$\begin{aligned} \text{Nodes}(\text{Leaf}) &= 1 \\ \text{Nodes}(\text{BNode}(T_1, T_2)) &= 1 + \text{Nodes}(T_1) + \text{Nodes}(T_2) \\ \text{Nodes}(\text{RNode}(T_1, T_2)) &= 1 + \text{Nodes}(T_1) + \text{Nodes}(T_2) \end{aligned}$$

Task 3. Prove by rule induction that for any tree T , color $c \in \{\text{Red}, \text{Black}\}$ and $n \geq 1$, if T isTree $c \ n$, then $\text{Nodes}(T) \geq 2^{n-1}$.

You must use rule induction, not any other proof technique.

Note: This, together with one or two other properties of red-black trees, proves that a valid red-black tree is approximately balanced.

Answer:

Proof: By rule induction on the derivation of T isTree $c \ n$. We have one case for each rule that could have concluded T isTree $c \ n$.

Case RBT-1: $T = \text{Leaf}$, $c = \text{Black}$, $n = 1$.

Case RBT-2: $T = \text{BNode}(T_1, T_2)$, $c = \text{Black}$, T derived from T_1 isTree $c_1 \ m$ and T_2 isTree $c_2 \ m$, with $n = m + 1$.

Case RBT-3: $T = \text{RNode}(T_1, T_2)$, $c = \text{Red}$, T derived from T_1 isTree Black n and T_2 isTree Black n . ■

4 One more wrap-up question

Task 4. How long (approximately) did you spend on this homework, in total minutes/hours of actual working time? Your honest feedback will help me with future homeworks.

Answer: I used AI to breakdown the problems and fix my latex code. I spent about 4 hours on this homework.